EA-101 Evacuation analysis (simplified method, MSC.1/Circ.1533)

SYN-1 (synthetic cruise ship) Yard hull S-701

SYNTHETIC TEST DATA: not a real ship

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1. Scope

This report gives the evacuation analysis of the ship required by SOLAS II-2/13.3.2.7. It uses the simplified method of MSC.1/Circ.1533. It covers the whole ship and both assessment cases, night and day.

2. Ship data

Persons on board (certified)	4,750
Passengers	3,300
Crew	1,450
Main vertical zones	7

3. Method

The total evacuation time is found from the response and travel time R+T and the embarkation and launching time E+L:

Total = 1.25 (R+T) + 2/3 (E+L)

4. Results

Case	R+T (min)	E+L (min)	Total (min)
Case 1 (night)	38.0	30.0	67.5
Case 2 (day)	35.0	29.0	63.1

Governing case: Case 1 (night)

Worked result for Case 1 (night): 1.25 x 38.0 + 2/3 x 30.0 = 47.5 + 20.0 = 67.5 min.

5. Conclusion

The governing total evacuation time is 67.5 min, Case 1 (night). The result is to be compared with the criterion of SOLAS II-2/13.3.2.7 for a ship with this number of main vertical zones.